

THANGKAHUB

A Premium Digital Marketplace and Cultural Preservation Platform for Authentic Nepali Thangka Art

1. Executive Summary

ThangkaHub is a modern web-based platform designed to connect traditional Nepali Thangka artists with global art collectors, cultural enthusiasts, and spiritual practitioners. The platform aims to digitize, preserve, and commercialize authentic Thangka artwork while promoting Nepal's rich artistic heritage worldwide.

Unlike conventional e-commerce platforms, ThangkaHub focuses on authenticity verification, artist-centric storytelling, commission-based artwork creation, educational content, and cultural preservation. The system enables artists to showcase their portfolios, share their artistic journey, conduct live interactive sessions, and receive custom commission requests from international buyers.

The platform leverages modern web technologies including Next.js, React, TypeScript, Supabase, Cloudflare Stream, and LiveKit to provide a scalable, secure, and immersive user experience.

2. Background and Problem Statement

Thangka painting is one of Nepal's most valuable cultural and artistic traditions. Despite its global recognition, many Nepali artists face challenges in reaching international buyers.

Current problems include:

- Lack of a dedicated digital marketplace for authentic Thangka artwork.
- Difficulty in verifying artwork authenticity.
- Limited visibility for individual artists.
- Dependence on intermediaries and physical galleries.
- Lack of educational resources explaining Thangka symbolism.
- Difficulty for customers to commission personalized artwork.
- Minimal use of digital technologies in traditional art promotion.

As global interest in Buddhist art and Himalayan culture grows, there is a need for a platform that combines commerce, culture, education, and authenticity.

3. Project Objectives

Primary Objectives

- Create a global marketplace for authentic Nepali Thangka art.
- Provide direct exposure to artists.
- Enable secure and transparent artwork sales.
- Preserve and promote Nepali cultural heritage.
- Establish authenticity verification mechanisms.

Secondary Objectives

- Support custom artwork commissions.
 - Allow live interaction between artists and buyers.
 - Educate users about Buddhist symbolism and Thangka traditions.
 - Create a digital archive of artistic processes.
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4. Scope of the System

The system covers:

Artist Management

- Artist registration
- Artist profile creation
- Portfolio management
- Verification status

Artwork Management

- Artwork uploads
- Product categorization
- Story-driven descriptions
- Authenticity certificates

Customer Services

- Browsing and searching artwork
- Wishlist management
- Commission requests
- Inquiry management

Educational Features

- Cultural information
- Symbol interpretation
- Historical context
- Artist stories

Community Features

- Live artist sessions
 - Interactive discussions
 - Event participation
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5. Proposed Features

5.1 Artist-Centric Profiles

Each artist receives:

- Biography
- Experience information
- Portfolio
- Specialization
- Awards and recognitions
- Introductory video

Benefits:

- Builds trust
 - Encourages artist branding
 - Improves buyer engagement
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5.2 Authenticity Verification System

Every artwork receives:

- Unique Artwork ID
- Digital Certificate
- Artist Verification
- Creation Date
- Materials Information
- QR Code Verification

Benefits:

- Reduces fraud
 - Increases customer confidence
 - Supports premium pricing
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5.3 Story-Driven Product Pages

Each artwork includes:

- Historical significance
- Deity information
- Symbol interpretation
- Artist inspiration
- Cultural context

Benefits:

- Enhances emotional value
 - Educates buyers
 - Improves user engagement
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5.4 Advanced Search and Filtering

Users can search by:

- Artist
- Price
- Size
- Deity
- Style
- Completion Year
- Color Palette
- Handmade Status

Benefits:

- Better user experience
 - Faster artwork discovery
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5.5 Multi-Currency Support

Supported currencies:

- NPR
- USD
- EUR
- GBP
- AUD

Benefits:

- International accessibility
 - Simplified purchasing process
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5.6 Time-Lapse Creation Archive

Artists can upload:

- Sketch progress
- Coloring stages
- Detail work
- Gold application
- Final artwork

Benefits:

- Demonstrates authenticity
 - Increases perceived value
 - Preserves artistic process
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5.7 Live Artist Sessions

Artists can:

- Host live painting sessions
- Answer customer questions
- Conduct workshops
- Demonstrate techniques

Benefits:

- Community engagement
- Increased transparency

- Artist promotion
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5.8 Commission Marketplace

Customers can:

- Request custom artwork
- Define specifications
- Receive quotations
- Track project progress

Benefits:

- Personalized experiences
 - New revenue opportunities
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6. System Architecture

Frontend Layer

Technology:

- Next.js 15
- React 19
- TypeScript
- Tailwind CSS
- ShadCN UI

Responsibilities:

- User interface
 - SEO optimization
 - Dynamic rendering
 - Responsive design
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Backend Layer

Technology:

- Next.js Server Actions
- Route Handlers

Responsibilities:

- Business logic
 - API services
 - Authentication
 - Data processing
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Database Layer

Technology:

- Supabase PostgreSQL

Responsibilities:

- User management
 - Artwork storage
 - Commission tracking
 - Session management
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Storage Layer

Technology:

- Supabase Storage

Stores:

- Images
 - Certificates
 - Videos
 - Documents
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Video Streaming Layer

Technology:

- Cloudflare Stream

Used for:

- Artist introductions
- Time-lapse videos

- Educational content
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Live Communication Layer

Technology:

- LiveKit

Used for:

- Live artist sessions
 - Interactive workshops
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Email Service

Technology:

- Resend

Used for:

- Notifications
 - Verification emails
 - Session reminders
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7. Database Design

Major Entities:

- Users
- Artists
- Artworks
- Categories
- Certificates
- Creation Steps
- Commission Requests
- Commission Proposals
- Orders
- Live Sessions
- Messages
- Notifications

Relationships:

Artist → Artworks

Artwork → Certificate

Artwork → Creation Steps

Customer → Commission Request

Commission Request → Proposal

Live Session → Artist

8. Non-Functional Requirements

Performance

- Page load time below 3 seconds
- Optimized image delivery
- CDN integration

Security

- Role-based access control
- Secure authentication
- HTTPS encryption
- Row-Level Security

Scalability

- Cloud deployment
- Modular architecture
- Microservice-ready design

Availability

- 99.9% uptime target
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9. Expected Benefits

For Artists

- Global exposure
- Direct customer communication
- Increased income opportunities

For Customers

- Authentic artwork
- Educational resources
- Personalized commissions

For Nepal

- Cultural preservation
 - International promotion of heritage
 - Economic opportunities for artists
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10. Challenges and Limitations

Challenges

- Artist onboarding
- Authenticity verification
- International shipping
- Currency fluctuations
- High-quality media storage

Limitations

- Internet dependency
 - Initial user acquisition
 - Payment gateway constraints
 - Live session scheduling
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11. Future Enhancements

- AI Cultural Guide
- AI Artwork Recognition

- Virtual Museum
 - Mobile Application
 - Auction Marketplace
 - Blockchain-based Authenticity Verification
 - AR Artwork Visualization
 - International Logistics Integration
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12. Conclusion

ThangkaHub aims to become a comprehensive digital ecosystem for Nepali Thangka art by combining modern technology with cultural preservation. Through artist empowerment, authenticity verification, educational storytelling, and community engagement, the platform creates value for artists, collectors, and the broader cultural heritage sector.

The project demonstrates how technology can be utilized not only for commerce but also for preserving and promoting traditional art forms on a global scale.